

# SMART INDIA HACKATHON 2026



## TITLE PAGE

Problem Statement ID – 26057

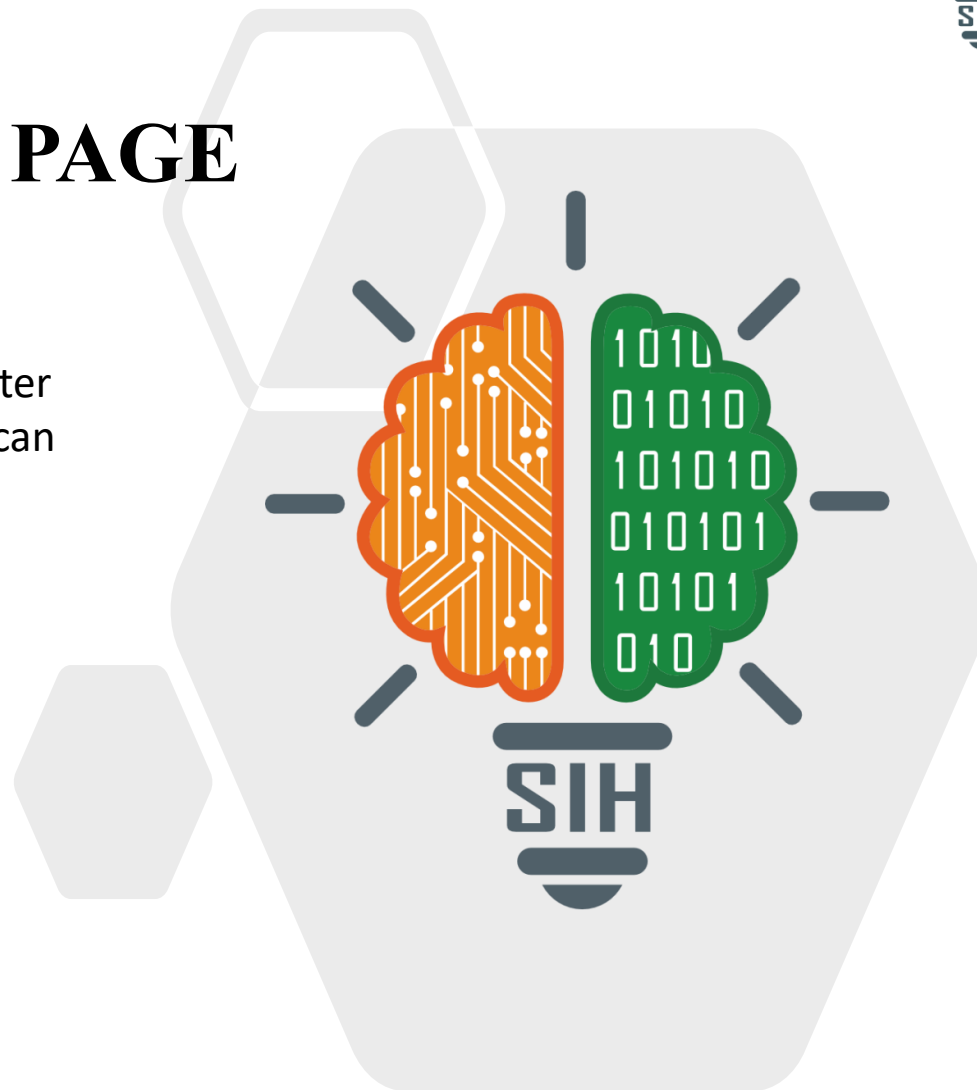
Problem Statement Title- AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery

Theme- Disaster Management

PS Category- Software

Team ID- <Team ID>

Team Name (Registered on portal)- <Team Name>



Your  
Team  
Name

# IDEA TITLE: SonarSentinel



## ❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution

### Proposed solution

SonarSentinel is an end-to-end AI pipeline + map dashboard that reads raw side-scan sonar logs (.xtf, GeoTIFF, image + navigation CSV), detects man-made debris and ghost nets, and reports each hazard's GPS position, size, class and a 0–100% confidence.

### How it addresses the problem

- Replaces slow manual review of sonar logs with automatic detection shown live on a map
- Separates debris from rocks, sand ripples and acoustic shadows using sonar physics + AI
- Handles speckle noise, varying pixel resolution, data dropouts and heave / pitch / roll
- Runs offline on a laptop or on an AUV / marine-drone edge computer

### Innovation and uniqueness

- GPS computed from sonar ping headers: every detection geotagged with an uncertainty estimate
- Finds ghost nets without a ghost-net dataset: synthetic nets + anomaly detection + expert review loop
- Physics-aware, calibrated and explainable confidence (shadow-geometry check + score breakdown)
- Edge-ready (INT8 TensorRT / OpenVINO) with open outputs: JSON, CSV, GeoJSON, KML

### Example output (illustrative)

Class: ghost net

Confidence: 87.4% · HAZARD

Position: 13.084120 N, 80.312750 E

Size: 6.2 m × 3.1 m · Depth: 18.5 m

Position uncertainty:  $\pm 4.2$  m

Source: line\_07.xtf, starboard side, pings 10,398–10,466

Delivered as JSON / CSV / KML for dive and ROV teams

### Workflow

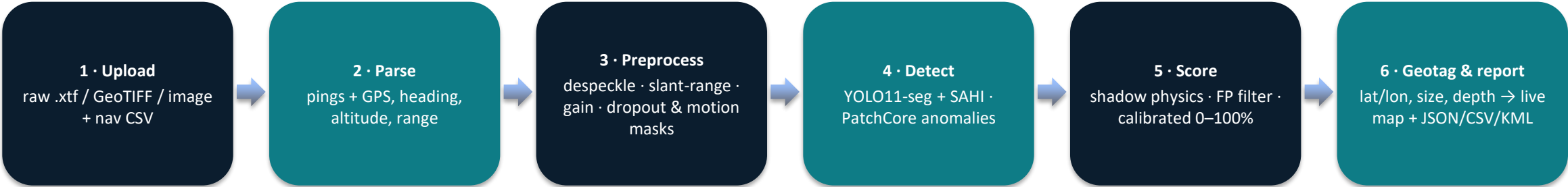
Upload sonar log → Detect → Score → Geotag → Live map + report

# TECHNICAL APPROACH

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

| Layer             | Technologies to be used                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Sonar I/O & geo   | Python 3.11 · pyxtf (XTF ping headers) · GDAL / rasterio · pyproj                                                                      |
| Signal processing | OpenCV · NumPy / SciPy: Lee despeckle, slant-range correction, gain normalisation, dropout & motion masks                              |
| AI models         | YOLO11-seg + SAHI (detection & masks) · PatchCore / anomalib (unknown objects) · LightGBM false-positive filter · isotonic calibration |
| Application       | FastAPI (REST + WebSocket live streaming) · React + Leaflet (offline maps) · SQLite → PostGIS                                          |
| Edge & deployment | ONNX · TensorRT (NVIDIA Jetson) · OpenVINO INT8 · Docker                                                                               |

## Methodology and process for implementation



- Long sonar logs are processed in overlapping chunks and streamed to the map in real time (WebSocket)
- Expert confirm / reject decisions become new training data; models are exported to ONNX for edge devices

- *Analysis of the feasibility of the idea*
- *Potential challenges and risks*
- *Strategies for overcoming these challenges*

### Analysis of feasibility

- Fully open-source stack (AGPL-3.0): runs on a GPU laptop, CPU-only, or a Jetson-class edge computer
- Public data available now: AI4Shipwrecks (286 labelled SSS images), mine-detection SSS dataset (1,170 images, CC BY 4.0), NOAA hydrographic surveys with navigation
- Six-sprint build plan with weekly milestones; requirements, architecture and test plan already defined

| Potential challenges and risks                       | Strategies for overcoming them                                                                                   |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| No public labelled ghost-net sonar dataset           | Synthetic ghost-net generator on real seabed + anomaly detection + expert review queue that collects real labels |
| Rocks, sand ripples and shadows cause false alarms   | Shadow-geometry check, shape / texture false-positive filter, hard-negative training, calibrated alert tiers     |
| Different sonars, frequencies and resolutions        | Resample to a fixed 10 cm / pixel, gain normalisation, sonar-specific augmentation, fine-tuning on NIOT samples  |
| Position errors (towfish layback, heading, GPS gaps) | Layback correction, navigation smoothing, per-detection uncertainty, validation against charted wrecks           |
| Limited compute on marine drones                     | Compact YOLO11 model with INT8 TensorRT / OpenVINO; heavy steps (anomaly model, mosaic) run on shore             |

# IMPACT AND BENEFITS

- *Potential impact on the target audience*
- *Benefits of the solution (social, economic, environmental, etc.)*

## Potential impact on the target audience

- NIOT / MoES survey teams: automated screening of survey logs; analysts review only flagged hazards
- Port, coastal and disaster authorities (Navy, Coast Guard): rapid post-cyclone / post-tsunami hazard maps
- Fisheries departments, NGOs and dive / ROV teams: exact GPS positions for ghost-net recovery

| Benefit              | Details                                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental        | Faster ghost-net discovery → less entanglement of fish, turtles and dolphins and less reef damage; supports UN SDG 14 (Life Below Water)    |
| Disaster management  | Rapid mapping of submerged debris in ports and shipping channels after cyclones, floods and tsunamis                                        |
| Economic             | Fewer analyst hours per survey (target ≥ 70% time saving, to be measured); less propeller and fishing-gear damage; faster channel reopening |
| Social & operational | Same-day hazard reports with GPS for field teams; on-board alerts; offline and on-premise, so sensitive seabed data stays in-country        |
| Strategic            | Indigenous, open, edge-deployable capability aligned with MoES / NIOT ocean-technology programmes                                           |

# RESEARCH AND REFERENCES



• *Details / Links of the reference and research work*

- [1] Z. Miao et al., "GhostNetZero: AI for detecting marine ghost nets," Microsoft AI for Good Lab, 2025.
- [2] A. V. Sethuraman et al., "Machine learning for shipwreck segmentation from side scan sonar imagery: Dataset and benchmark," Int. J. Robotics Research, 44(3), 2025. doi:10.1177/02783649241266853
- [3] N. Pessanha Santos et al., "Side-scan sonar imaging data of underwater vehicles for mine detection," Data in Brief, 53, 2024. doi:10.1016/j.dib.2024.110132
- [4] G. Huo, Z. Wu, J. Li, "Underwater object classification in sidescan sonar images using deep transfer learning and semisynthetic training data," IEEE Access, 8, 2020.
- [5] K. Roth et al., "Towards total recall in industrial anomaly detection" (PatchCore), CVPR 2022.
- [6] F. C. Akyon, S. O. Altinuc, A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," ICIP 2022.
- [7] C. Guo et al., "On calibration of modern neural networks," ICML 2017.
- [8] K. Richardson, B. D. Hardesty, C. Wilcox, "Estimates of fishing gear loss rates at a global scale," Fish and Fisheries, 20(6), 2019. doi:10.1111/faf.12407
- [9] P. Blondel, The Handbook of Sidescan Sonar, Springer/Praxis, 2009.
- [10] Data & tools: NOAA NCEI hydrographic survey data; Ultralytics YOLO11; pyxtf (github.com/oysstu/pyxtf); anomalib.